MATH 342W / 642 / RM742 Spring 2026 HW #2

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Problem 1

These are questions about Silver's book, chapter 2, 3. Answer the questions using notation from class (i.e. $t, f, g, h^*, \delta, \epsilon, e, t, z_1, \ldots, z_t, \mathbb{D}, \mathcal{H}, \mathcal{A}, \mathcal{X}, \mathcal{Y}, X, y, n, p, x_{\cdot 1}, \ldots, x_{\cdot p}, x_{1\cdot}, \ldots, x_{n\cdot}$, etc).

(a) [harder] If one's goal is to fit a model for a phenomenon y, what is the difference between the approaches of the hedgehog and the fox? Connecting this to the modeling framework should really make you think about what Tetlock's observation means for political and historical phenomena.

> If we are trying to model a phenomenon $y = t(z_1, z_2, ..., z_t)$, a hedgehog will believe in one outcome of t, not changing their model when new data comes in. A fox is more adaptable, updating their prediction g and models h as new data arrives.
> Because of this, foxes tend to be better forecasters.

(b) [easy] Why did Harry Truman like hedgehogs? Are there a lot of people that think this way?

> He preferred simpler, more concrete answers as opposed to more complicated, uncertainty filled answers that foxes give. This is a sentiment that many people share, preferring the simplicity of hedgehog answers.

(c) [difficult] Why is it that the more education one acquires, the less accurate one's predictions become?

> According to Tetlock, when one acquires more education, it allows them to use that knowledge to manipulate data, finding ways to fit that data into their predictions in a way that confirms their own theories. This can lead to increased prediction error.

(d) [easy] Why are probabilistic classifiers (i.e. algorithms that output functions that return probabilities) better than vanilla classifiers (i.e. algorithms that only return the class label)? We will move in this direction in class soon.

> Instead of outputting a single number, i.e 1 or 0, yes or no, right or wrong, probabilistic classifiers can tell you how confident a model is in its prediction. A "yes" with a confidence of 58% is much more valuable than simply "yes".

(e) [easy] What algorithm that we studied in class is PECOTA most similar to?

> PECOTA is most similar to the KNN (nearest neighbor) algorithm, comparing players to others in order to generate a prediction for their careers and abilites.

(f) [easy] Is baseball performance as a function of age a linear model? Discuss.

> No, players tend to increase, peak, and then decline.

(g) [harder] How can baseball scouts do better than a prediction system like PECOTA?

> The scouts have access to information outside of statistics, so are able to notice things that are not in the data PECOTA has access to. They can find better proxies for the z's, like using radar guns to measure fastballs. This reduces ignorance error δ

(h) [harder] Why hasn't anyone (at the time of the writing of Silver's book) taken advantage of Pitch f/x data to predict future success?

> Because the data from Pitch f/x was hard to model, with complicated and more arbitrary statistics, like best slider.

Problem 2

These are questions about the SVM.

(a) [easy] State the hypothesis set $\mathcal{H}$ inputted into the support vector machine algorithm. Is it different than the $\mathcal{H}$ used for $\mathcal{A}$ = perceptron learning algorithm?

> The hypothesis set for $\mathcal{H}$ is $\mathcal{H} = \{\mathbb{1}_{\vec{w}\cdot\vec{x}-b\geq 0} : \vec{w} \in \mathbb{R}^p, b \in \mathbb{R}\}$. This is the same hypothesis set used for the perceptron learning algorithm.

(b) [E.C.] Prove the max-margin linearly separable SVM converges. State all assumptions. Write it on a separate page.

(c) [difficult] Let $\mathcal{Y} = \{-1, 1\}$. Rederive the cost function whose minimization yields the SVM line in the linearly separable case.

> $\ell_U : \vec{w}'x - (b' + 1) = 0$ and $\ell_L : \vec{w}'x - (b' - 1) = 0$
>
> We wish to maximize m and minimize $||w'||$ subject to these constraints:
> 1. $\forall i$ such that $y_1 = 1$, the $\vec{x_i}$'s must be above ℓ_U
> $\Rightarrow \vec{w}'x_i - (b' + 1) \geq 0$
> $\Rightarrow \vec{w}'x_i - b' \geq 1$
> $y_i = 1$, so $y_1(\vec{w}'x_i - b') \geq 1$
>
> 2. 1. $\forall i$ such that $y_1 = -1$, the $\vec{x_i}$'s must be below ℓ_L
> $\Rightarrow \vec{w}'x_i - (b' - 1) \leq 0$
> $\Rightarrow \vec{w}'x_i - b' \leq -1$
> Multiply by -1 $\Rightarrow -1(\vec{w}'x_i - b') \geq 1 \Rightarrow y_i = -1$, so $y_1(\vec{w}'x_i - b') \geq 1$
> There is only one constraint: $\forall i\ y_1(\vec{w}'x_i - b') \geq 1$

(d) [easy] Given your answer to (c) rederive the cost function using the "soft margin" i.e. the hinge loss plus the term with the hyperparameter λ. This is marked easy since there is just one change from the expression given in class.

> Let H_i be the "hinge error" for observation i.
>
> $H_i = \max\{0, , 1 - y_1(\vec{w}'x_i - b')\}$
> Let $SHE := \sum_{i-1}^{n} H_i$
>
> The soft margin is $A : argmin\{\frac{1}{n}SHE + \lambda||\vec{w}'||^2\}$ where $\lambda \geq 0$

Problem 3

These are questions are about the k nearest neighbors (KNN) algorithm.

(a) [easy] Describe how the algorithm works. Is k a "hyperparameter"?

> The KNN algorithm works by analyzing the dataset and finding the most closest match by calculating the smallest y_i such that $d(\vec{x_i}, \vec{x_n})$ is the smallest for all $i = 1, 2, ..., n$.
> k is a hyperparameter, we choose it before running the algorithm.

(b) [difficult] [MA] Assuming $\mathcal{A}$ = KNN, describe the input $\mathcal{H}$ as best as you can.

> $\mathcal{A}$ is an algorithm using $\mathcal{D}$ and $\mathcal{H}$: $\mathcal{A} = (\mathcal{D}, \mathcal{H})$, where $\mathcal{H}$ is the set of candidate functions. So for KNN, $\mathcal{H}$ is the set of functions that looks at the input x and looks at the k nearest numbers in $\mathcal{D}$ and returns the closest point on average.

(c) [easy] When predicting on $\mathbb{D}$ with $k = 1$, why should there be zero error? Is this a good estimate of future error when new data comes in? (Error in the future is called *generalization error* and we will be discussing this later in the semester).

> In this scenario, the closest neighbor to each point is itself, meaning there is no error in the model.
> Because the prediction is controlled by only one point, any small changes in the data can change the prediction, meaning the error could change drastically. So it is not a good estimate of future error.

Problem 4

These are questions about the linear model with $p = 1$.

(a) [easy] What does $\mathbb{D}$ look like in the linear model with $p = 1$? What is $\mathcal{X}$? What is $\mathcal{Y}$?

> $\mathcal{D} = \{< x_1, y_1 >, ..., < x_n, y_n >\}$ where each input is a single real number.
> $\mathcal{X} \subseteq \mathbb{R}$ and $\mathcal{Y} \subseteq \mathbb{R}$

(b) [easy] Consider the line fit using the ordinary least squares (OLS) algorithm. Prove that the point $< \bar{x}, \bar{y} >$ is on this line. Use the formulas we derived in class.

> $g(x) = b_0 + b_1 x$ or $g(\overline{x}) = b_0 + b_1 \overline{x}$
> We also know $b_0 = \overline{y} - b_1 \overline{x}$, so $g(\overline{x}) = (\overline{y} - b_1 \overline{x}) + b_1 \overline{x}$
> $\Rightarrow g(\overline{x}) = \overline{y}$

(c) [harder] Consider the line fit using OLS. Prove that the average prediction $\hat{y}_i := g(x_i)$ for $x_i \in \mathbb{D}$ is $\bar{y}$.

We want to prove that $\frac{1}{n}\sum_{i=1}^{n}\hat{y}_i = \bar{y}$ where $\hat{y}_i := g(x_i)$.
$g(x) = b_0 + b_1 x \Rightarrow \hat{y}_i = b_0 + b_1 x_i$
Average both sides: $\frac{1}{n}\sum_{i=1}^{n}\hat{y}_i = \frac{1}{n}\sum_{i=1}^{n}(b_0+b_1x_i) \Rightarrow (b_0+b_1)\frac{1}{n}\sum_{i=1}^{n}x_i = b_0+b_1\hat{x}$
$b_0 = \overline{y} - b_1\overline{x}$ so $b_0 + b_1\hat{x} = (\overline{y} - b_1\overline{x}) + b_1\overline{x} = \hat{y}$
Therefore $\frac{1}{n}\sum_{i=1}^{n}\hat{y}_i = \bar{y}$

(d) [harder] Consider the line fit using OLS. Prove that the average residual e_i is 0 over $\mathbb{D}$.

$g(x) = b_0 + b_1 x \Rightarrow \hat{y}_i = b_0 + b_1 x_i$
$e_i := y_i - \hat{y}_i$
Average both sides: $\frac{1}{n}\sum_{i=1}^{n}e_i = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i) = \overline{y} - \frac{1}{n}\sum_{i=1}^{n}\hat{y}_i$
From part c: $\frac{1}{n}\sum_{i=1}^{n}\hat{y}_i = \bar{y}$
So: $\frac{1}{n}\sum_{i=1}^{n}e_i = \overline{y} - \overline{y} = 0$

(e) [harder] Why is the RMSE usually a better indicator of predictive performance than R^2? Discuss in English.

RMSE tells us how large our predictions errors are, measured in units of y^2. R^2 gives us the proportion of variation relative to $\overline{y}$.

(f) [harder] R^2 is commonly interpreted as "proportion of the variance explained by the model" and proportions are constrained to the interval $[0, 1]$. While it is true that $R^2 \leq 1$ for all models, it is not true that $R^2 \geq 0$ for all models. Construct an explicit example $\mathbb{D}$ and create a linear model $g(x) = w_0 + w_1 x$ whose $R^2 < 0$.

The equation for R^2 is $1 - \frac{SSE}{SST}$, so in order to have $R^2 < 0$, $\frac{SSE}{SST}$ must be more than 1: $SSE > SST$.
$\mathbb{D} = \{<0,0>, <1,1>\} \Rightarrow \overline{y} = \frac{1}{2}$
$SST = (0-\frac{1}{2})^2 + (1-\frac{1}{2})^2 = \frac{1}{2}$
Make model $g(x) = 10$
$SSE = (0-10)^2 + (1-10)^2 = 181$
Now $R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{181}{1/2}$,$R^2 < 0$

(g) [difficult] You are given $\mathbb{D}$ with n training points $< x_i, y_i >$ but now you are also given a set of weights $[w_1\ w_2\ \ldots\ w_n]$ which indicate how costly the error is for each of the i points. Rederive the least squares estimates b_0 and b_1 under this situation. Note that these estimates are called the *weighted least squares regression* estimates. This variant $\mathcal{A}$ on OLS has a number of practical uses, especially in Economics. No need to simplify your answers like I did in class (i.e. you can leave in ugly sums).

We minimize the weighted sum of squared errors: $\frac{1}{n}\sum_{i=1}^{n} w_i(y_i - b_0 - b_1 x_i)^2$
Take the partial derivitives and set them equal to 0:
$\frac{\partial}{\partial b_0}\sum_{i=1}^{n} w_i(y_i - b_0 - b_1 x_i)^2 = -2\sum_{i=1}^{n} w_i(y_i - b_0 - b_1 x_i) = 0$
$\frac{\partial}{\partial b_1}\sum_{i=1}^{n} w_i(y_i - b_0 - b_1 x_i)^2 = -2\sum_{i=1}^{n} w_i x_i(y_i - b_0 - b_1 x_i) = 0$

Rearrange to get:
$\sum_{i=1}^{n} w_i y_i = b_0\sum_{i=1}^{n} w_i + b_1\sum_{i=1}^{n} w_i x_i$ and
$\sum_{i=1}^{n} w_i x_i y_i = b_0\sum_{i=1}^{n} w_i x_i + b_1\sum_{i=1}^{n} w_i x_i^2$

$S_w = \sum_{i=1}^{n} w_i,\ S_x = \sum_{i=1}^{n} w_i x_i,\ S_{xx} = \sum_{i=1}^{n} w_i x_i^2,\ S_{xy} = \sum_{i=1}^{n} w_i x_i y_i,\ S_y = \sum_{i=1}^{n} w_i y_i$

$b_1 = \frac{S_w S_{xy} - S_x S_y}{S_w S_{xx} - S_x^2}$
$b_0 = \frac{S_y - b_1 S_x}{S_w}$
(used ChatGPT)

(h) [harder] Interpret the ugly sums in the b_0 and b_1 you derived above and compare them to the b_0 and b_1 estimates in OLS. Does it make sense each term should be altered in this matter given your goal in the weighted least squares?

In OLS, the observations have the same weight, the sums in b_0 and b_1 treat every observation with equal importance. So the equations from above would become:
$\sum_{i=1}^{n} w_i x_i = \sum_{i=1}^{n} x_i$
$\sum_{i=1}^{n} w_i x_i = \sum_{i=1}^{n} y_i$
$\sum_{i=1}^{n} w_i x_i^2 = \sum_{i=1}^{n} x_i^2$
$\sum_{i=1}^{n} w_i x_i^2 = \sum_{i=1}^{n} x_i y_i$
and $\sum_{i=1}^{n} w_i = n$
In the WLS version, each of those sums gets weighted by w_i. Observations with more weight will impact the fit more than lower weight observations. This makes sense because we want to make sure the observations that have larger weights, and therefore more influential error, are adjusted properly.

(i) [E.C.] In class we talked about $x_{raw} \in \{\text{red}, \text{green}\}$ and the OLS model was the sample average of the inputted x. Imagine if you have the additional constraint that x_{raw} is ordinal e.g. $x_{raw} \in \{\text{low}, \text{high}\}$ and you were forced to have a model where $g(\text{low}) \leq g(\text{high})$. Write about an algorithm $\mathcal{A}$ that can solve this problem.

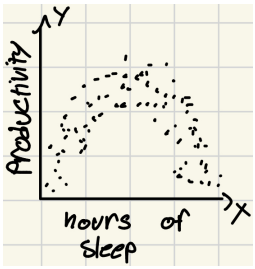
Problem 5

These are questions about association and correlation.

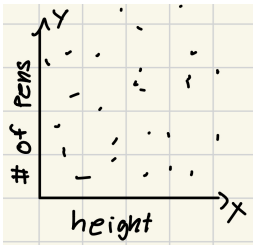
(a) [easy] Give an example of two variables that are both correlated and associated by drawing a plot.



(b) [easy] Give an example of two variables that are not correlated but are associated by drawing a plot.



(c) [easy] Give an example of two variables that are not correlated nor associated by drawing a plot.



(d) [easy] Can two variables be correlated but not associated? Explain.

> If two variables are correlated, that means there is a linear relationship, meaning it must also be associated.

Problem 6

These are questions about multivariate linear model fitting using the least squares algorithm.

(a) [difficult] Derive $\frac{\partial}{\partial \boldsymbol{c}}\left[\boldsymbol{c}^\top A \boldsymbol{c}\right]$ where $\boldsymbol{c} \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n \times n}$ but *not* symmetric. Get as far as you can.

(b) [easy] Given matrix $X \in \mathbb{R}^{n \times (p+1)}$, full rank and first column consisting of the $\mathbf{1}_n$ vector, rederive the least squares solution $\boldsymbol{b}$ (the vector of coefficients in the linear model shipped in the prediction function g). No need to rederive the facts about vector derivatives.

(c) [harder] Consider the case where $p = 1$. Show that the solution for $\boldsymbol{b}$ you just derived in (b) is the same solution that we proved for simple regression. That is, the first element of $\boldsymbol{b}$ is the same as $b_0 = \bar{y} - r\frac{s_y}{s_x}\bar{x}$ and the second element of $\boldsymbol{b}$ is $b_1 = r\frac{s_y}{s_x}$.

(d) [easy] If X is rank deficient, how can you solve for $\boldsymbol{b}$? Explain in English.

(e) [difficult] Prove $\text{rank}\,[X] = \text{rank}\left[X^\top X\right]$.

(f) [harder] [MA] If $p = 1$, prove $r^2 = R^2$ i.e. the linear correlation is the same as proportion of sample variance explained in a least squares linear model.

(g) [harder] Prove that $g([1\ \bar{x}_1\ \bar{x}_2\ \ldots\ \bar{x}_p]) = \bar{y}$ in OLS.

(h) [harder] Prove that $\bar{e} = 0$ in OLS.

(i) [difficult] If you model $\boldsymbol{y}$ with one categorical nominal variable that has levels A, B, C, prove that the OLS estimates look like $\bar{y}_A$ if $x = A$, $\bar{y}_B$ if $x = B$ and $\bar{y}_C$ if $x = C$. You can choose to use an intercept or not. Likely without is easier.

(j) [harder] [MA] Prove that the OLS model always has $R^2 \in [0, 1]$.